

Having now fixed T , we choose δ appropriately small. On γ_3 , note that

$$|x^{1+s}| = x^{1+1-\delta} = x^{2-\delta},$$

from which we conclude that there exists a constant C_T (dependent on T) such that

$$\left| \int_{\gamma_3} F(s) ds \right| \leq C_T x^{2-\delta}.$$

Finally, on the small horizontal segment γ_2 (and similarly on γ_4), we can estimate the integral as follows:

$$\left| \int_{\gamma_2} F(s) ds \right| \leq C'_T \int_{1-\delta}^1 x^{1+\sigma} d\sigma \leq C'_T \frac{x^2}{\log x}.$$

We conclude that there exist constants C_T and C'_T (possibly different from the ones above) such that

$$\left| \psi_1(x) - \frac{x^2}{2} \right| \leq \epsilon x^2 + C_T x^{2-\delta} + C'_T \frac{x^2}{\log x}.$$

Dividing through by $x^2/2$, we see that

$$\left| \frac{2\psi_1(x)}{x^2} - 1 \right| \leq 2\epsilon + 2C_T x^{-\delta} + 2C'_T \frac{1}{\log x},$$

and therefore, for all large x we have

$$\left| \frac{2\psi_1(x)}{x^2} - 1 \right| \leq 4\epsilon.$$

This concludes the proof that

$$\psi_1(x) \sim x^2/2 \quad \text{as } x \rightarrow \infty,$$

and thus, we have also completed the proof of the prime number theorem.

Note on interchanging double sums

We prove the following facts about the interchange of infinite sums: if $\{a_{k\ell}\}_{1 \leq k, \ell < \infty}$ is a sequence of complex numbers indexed by $\mathbb{N} \times \mathbb{N}$, such that

$$(8) \quad \sum_{k=1}^{\infty} \left(\sum_{\ell=1}^{\infty} |a_{k\ell}| \right) < \infty,$$

then:

- (i) The double sum $A = \sum_{k=1}^{\infty} (\sum_{\ell=1}^{\infty} a_{k\ell})$ summed in this order converges, and we may in fact also interchange the order of summation, so that

$$A = \sum_{k=1}^{\infty} \sum_{\ell=1}^{\infty} a_{k\ell} = \sum_{\ell=1}^{\infty} \sum_{k=1}^{\infty} a_{k\ell}.$$

- (ii) Given $\epsilon > 0$, there is a positive integer N so that for all $K, L > N$ we have
- $$\left| A - \sum_{k=1}^K \sum_{\ell=1}^L a_{k\ell} \right| < \epsilon.$$

- (iii) If $m \mapsto (k(m), \ell(m))$ is a bijection from $\mathbb{N}$ to $\mathbb{N} \times \mathbb{N}$, and if we write $c_m = a_{k(m)\ell(m)}$, then $A = \sum_{k=1}^{\infty} c_k$.

Statement (iii) says that any rearrangement of the sequence $\{a_{k\ell}\}$ can be summed without changing the limit. This is analogous to the case of absolutely convergent series, which can be summed in any desired order.

The condition (8) says that each sum $\sum_{\ell} a_{k\ell}$ converges absolutely, and moreover this convergence is “uniform” in k . An analogous situation arises for sequences of functions, where an important question is whether or not the interchange of limits

$$\lim_{x \rightarrow x_0} \lim_{n \rightarrow \infty} f_n(x) \stackrel{?}{=} \lim_{n \rightarrow \infty} \lim_{x \rightarrow x_0} f_n(x)$$

holds. It is a well-known fact that if the f_n 's are continuous, and their convergence is uniform, then the above identity is true since the limit function is itself continuous. To take advantage of this fact, define $b_k = \sum_{\ell=1}^{\infty} |a_{k\ell}|$ and let $S = \{x_0, x_1, \dots\}$ be a countable set of points with $\lim_{n \rightarrow \infty} x_n = x_0$. Also, define functions on S as follows:

$$\begin{aligned} f_k(x_0) &= \sum_{\ell=1}^{\infty} a_{k\ell} && \text{for } k = 1, 2, \dots \\ f_k(x_n) &= \sum_{\ell=1}^n a_{k\ell} && \text{for } k = 1, 2, \dots \text{ and } n = 1, 2, \dots \\ g(x) &= \sum_{k=1}^{\infty} f_k(x) && \text{for } x \in S. \end{aligned}$$

By assumption (8), each f_k is continuous at x_0 . Moreover $|f_k(x)| \leq b_k$ and $\sum b_k < \infty$, so the series defining the function g is uniformly convergent on S , and therefore g is also continuous at x_0 . As a consequence we find (i), since

$$\begin{aligned} \sum_{k=1}^{\infty} \sum_{\ell=1}^{\infty} a_{k\ell} &= g(x_0) = \lim_{n \rightarrow \infty} g(x_n) = \lim_{n \rightarrow \infty} \sum_{k=1}^{\infty} \sum_{\ell=1}^n a_{k\ell} \\ &= \lim_{n \rightarrow \infty} \sum_{\ell=1}^n \sum_{k=1}^{\infty} a_{k\ell} = \sum_{\ell=1}^{\infty} \sum_{k=1}^{\infty} a_{k\ell}. \end{aligned}$$

For the second statement, first observe that

$$\left| A - \sum_{k=1}^K \sum_{\ell=1}^L a_{k\ell} \right| \leq \sum_{k \leq K} \sum_{\ell > L} |a_{k\ell}| + \sum_{k > K} \sum_{\ell=1}^{\infty} |a_{k\ell}|.$$

To estimate the second term, we use the fact that $\sum b_k$ converges, which implies $\sum_{k>K} \sum_{\ell=1}^{\infty} |a_{k\ell}| < \epsilon/2$ whenever $K > K_0$, for some K_0 . For the first term above, note that $\sum_{k \leq K} \sum_{\ell > L} |a_{k\ell}| \leq \sum_{k=1}^{\infty} \sum_{\ell > L} |a_{k\ell}|$. But the argument above guarantees that we can interchange these last two sums; also $\sum_{\ell=1}^{\infty} \sum_{k=1}^{\infty} |a_{k\ell}| < \infty$, so that for all $L > L_0$ we have $\sum_{\ell > L} \sum_{k=1}^{\infty} |a_{k\ell}| < \epsilon/2$. Taking $N > \max(L_0, K_0)$ completes the proof of (ii).

The proof of (iii) is a direct consequence of (ii). Indeed, given any rectangle

$$R(K, L) = \{(k, \ell) \in \mathbb{N} \times \mathbb{N} : 1 \leq k \leq K \text{ and } 1 \leq \ell \leq L\},$$

there exists M such that the image of $[1, M]$ under the map $m \mapsto (k(m), \ell(m))$ contains $R(K, L)$.

When U denotes any open set in $\mathbb{R}^2$ that contains the origin, we define for $R > 0$ its dilate $U(R) = \{y \in \mathbb{R}^2 : y = Rx \text{ for some } x \in U\}$, and we can apply (ii) to see that

$$A = \lim_{R \rightarrow \infty} \sum_{(k, \ell) \in U(R)} a_{k\ell}.$$

In other words, under condition (8) the double sum $\sum_{k\ell} a_{k\ell}$ can be evaluated by summing over discs, squares, rectangles, ellipses, etc.

Finally, we leave the reader with the instructive task of finding a sequence of complex numbers $\{a_{k\ell}\}$ such that

$$\sum_k \sum_{\ell} a_{k\ell} \neq \sum_{\ell} \sum_k a_{k\ell}.$$

[Hint: Consider $\{a_{k\ell}\}$ as the entries of an infinite matrix with 0 above the diagonal, -1 on the diagonal, and $a_{k\ell} = 2^{\ell-k}$ if $k > \ell$.]

3 Exercises

1. Suppose that $\{a_n\}_{n=1}^{\infty}$ is a sequence of real numbers such that the partial sums

$$A_n = a_1 + \cdots + a_n$$

are bounded. Prove that the Dirichlet series

$$\sum_{n=1}^{\infty} \frac{a_n}{n^s}$$

converges for $\operatorname{Re}(s) > 0$ and defines a holomorphic function in this half-plane.

[Hint: Use summation by parts to compare the original (non-absolutely convergent) series to the (absolutely convergent) series $\sum A_n(n^{-s} - (n+1)^{-s})$. An estimate for the term in parentheses is provided by the mean value theorem. To prove that the series is analytic, show that the partial sums converge uniformly on every compact subset of the half-plane $\operatorname{Re}(s) > 0$.]